

Homework Classical Physics, Fall 2023

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Problem Set 04.2 (L07 & L08)

Due on Thursday, 5. October 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - Work and energy (4 points)

Sisyphos is pushing, as usual, a rock of mass $m = 50 \text{ kg}$ with constant speed uphill on a slope of 20° . Unfortunately he also has to overcome a frictional coefficient of $\mu_{kin} = 0.20$.

- a) Calculate the work he has done after pushing it a distance $s = 100 \text{ m}$.
- b) Calculate the gain in potential energy of the rock after the distance s .
- c) Explain the difference in energy between (a) and (b).